AI Macro Automated Refresh

GitHub + Streamlit Deployment and Operations Instructions

Use this document when the project is ready to be pushed to GitHub and connected or updated on Streamlit. The public app remains retained-only; scheduled GitHub Actions perform validated refreshes independently.

1. Required project files

Push the complete project to the repository's default branch. Confirm that it includes the workflow, refresh helpers, retained-only policy, snapshot writer, retained-data directories, the Streamlit app, and requirements files.

- .github/workflows/refresh-data.yml
- helpers/refresh_pipeline.py
- helpers/refresh_validate.py
- helpers/refresh_diff.py
- config/load_policy.py
- loaders/snapshot_writer.py
- data/ and archive/ retained-data directories
- The complete Streamlit application and requirements files

The workflow must exist on the default branch before normal manual workflow dispatch is available.

2. Add GitHub Actions secrets

1. Open the repository in GitHub.
2. Go to Settings -> Secrets and variables -> Actions.
3. Create FRED_API_KEY with a valid FRED API key.
4. Create SEC_USER_AGENT with a descriptive SEC identity and real contact address, for example: AI Macro Research your-email@example.com.

Do not place either secret in source code, workflow YAML, logs, or committed configuration files.

3. Enable workflow write permission

1. Open Settings -> Actions -> General.
2. Scroll to Workflow permissions.
3. Select Read and write permissions.
4. Save the setting.

```
permissions:
  contents: write
```

The repository setting and the workflow permission should both be present so accepted snapshots can be committed.

4. First deployment sequence

Do not rely on the schedules until every profile has completed one successful dry run and one successful publication run.

A. market_daily

1. Open Actions -> Refresh retained data -> Run workflow.
2. Choose market_daily and set dry_run to true.
3. Run it and inspect the uploaded refresh evidence.
4. Confirm YFinance, FRED, NY Fed, and Current Context all passed.
5. Confirm no candidate validation gate failed.
6. Run market_daily again with dry_run set to false.
7. Confirm one coherent retained-data commit was created.
8. Confirm data/refresh_manifest.json was created or updated.
9. Confirm the first successful FRED publication created or updated data/finance/nfci_anfci_history.csv and passed Section 10.

B. fundamentals_weekly

1. Run fundamentals_weekly with dry_run set to true.
2. Inspect the evidence and confirm EDGAR passed completeness and mode checks.
3. Run fundamentals_weekly with dry_run set to false.
4. Confirm the accepted retained-data update was committed.

C. domains_weekly

1. Run domains_weekly with dry_run set to true.
2. Inspect the evidence and confirm every requested domain passed.
3. Run domains_weekly with dry_run set to false.
4. Confirm the accepted retained-data update was committed.

Do not publish merely to create the first refresh manifest. A failed contract must be repaired or deliberately redefined, not bypassed.

5. Scheduled profiles

Profile	Schedule	Sources
market_daily	Weekdays 23:37 UTC	YFinance, FRED, NY Fed, Current Context
fundamentals_weekly	Saturday 15:17 UTC	EDGAR
domains_weekly	Sunday 16:43 UTC	Compute, Data Centers, Connectivity, Power, Grid & Storage, Water, Adoption, Workforce, Economic Outcomes, Current Context
full	Manual only	Every provider and domain
retained_check	Manual only; dry-run required	No network; validation only

GitHub schedules can be delayed. Freshness must use actual per-source success timestamps from the manifest.

6. Manual refresh and targeted recovery

1. Open Actions -> Refresh retained data -> Run workflow.

2. Choose the appropriate profile.
3. Start with dry_run set to true.
4. Inspect the uploaded evidence.
5. Publish only after the dry run passes.

For targeted recovery, use the optional sources input with comma-separated RefreshSource values, such as yfinance or compute,adoption.

Use a narrow source override to recover an unrelated healthy source. Do not weaken the failed source's acceptance contract.

7. Local dry-run command

```
AI_MACRO_MODE=developer \
FRED_API_KEY='your-key' \
SEC_USER_AGENT='AI Macro Research your-email@example.com' \
python helpers/refresh_pipeline.py --profile full --dry-run
```

Inspect audit/refresh_runs/. Remove --dry-run only after the evidence passes review.

8. Transaction and failure behavior

1. Copy the current project into a temporary staging directory.
2. Authorize live network access only for requested sources.
3. Perform loader writes only inside staging.
4. Reject incomplete provider results.
5. Parse every candidate retained file.
6. Compare the candidate against the published snapshot.
7. Reject deletion, schema loss, material row collapse, historical contraction, date regression, invalid files, and extreme unexpected growth.
8. Prepare rollback copies and promote the accepted candidate.
9. Install data/refresh_manifest.json last as the transaction marker.
10. Commit archive/ and data/ together.

If any requested source or gate fails, publish nothing from that profile. Preserve the previous validated snapshot and review the uploaded evidence.

Never manually copy staged files into the repository to bypass a failed gate.

9. Streamlit behavior

The public Streamlit application remains a retained-only reader. Viewer sessions must not contact providers, run discovery refreshes, write retained data, or publish snapshots.

- Successful GitHub Actions commits deliver validated snapshots to Streamlit.
- Viewer startup performs zero provider network calls.
- Viewer startup performs zero retained-data writes.
- The validated snapshot date is displayed.
- Stale-source warnings use per-source timestamps.
- Weekly domain updates do not make market data appear newer.
- Only a successful YFinance transaction may advance the top-level market date.

10. NFCI / ANFCI historical acceptance gate

The dedicated retained Finance history is data/finance/nfci_anfci_history.csv.

- At least 500 valid NFCI observations.
- At least 500 valid ANFCI observations.
- At least 3,560 days of historical span.
- A valid latest observation.
- No contraction of the previously published historical window.

archive/fred_history.csv is a latest-snapshot archive and must not replace the dedicated historical series.

Until the first successful FRED publication passes this gate, the bundled short fallback remains incomplete and the Finance chart must not be treated as repaired.

11. Success checklist after each publish

- The workflow completed successfully.
- The refresh evidence artifact was uploaded.
- Every requested source passed.
- No unrelated source was authorized.
- One coherent retained-data commit was created.
- archive/ and data/ were committed together when changed.
- data/refresh_manifest.json contains the new snapshot identifier.
- Per-source success timestamps are correct.
- Observation and release dates did not regress.
- Streamlit shows the new validated snapshot.
- Streamlit startup remains fast and retained-only.
- No viewer-triggered provider calls occur.

12. Ongoing operations

Review GitHub Actions for failed schedules, credential failures, provider mode failures, coverage reductions, historical-range failures, schema changes, file-growth anomalies, date regressions, and repeated stale-source warnings.

Failure artifacts are configured for 30-day retention. Download important evidence before it expires.

When a scheduled run fails, keep serving the previous validated snapshot and investigate before attempting publication again.

13. Do not do these things

- Do not let Streamlit viewer sessions refresh providers.
- Do not interpret cache clearing as refresh authorization.
- Do not publish retained fallbacks as fresh live results.
- Do not bypass failed validation by copying staged files manually.
- Do not combine an unrelated code release with an unreviewed data refresh.
- Do not commit secrets.

- Do not let domain refreshes advance the market date.
- Do not replace historical series with latest-snapshot archives.
- Do not run retained_check without dry-run.
- Do not trust the first green live workflow without reviewing its evidence.

14. Recommended release practice

1. Keep the automated-refresh build and these instructions stored safely.
2. Continue development on a separate branch or working copy.
3. Before the eventual GitHub push, merge or reapply the automated-refresh commit carefully.
4. Run the full packaged test suite and integrity gate after the merge.
5. Confirm .github/workflows/refresh-data.yml is still present.
6. Confirm retained-only startup has not regressed.
7. Complete the first-deployment dry-run sequence in Section 4.

```
Branch: recovery/v6.9.1-automated-refresh
Commit: 1a288ca3077d8413349f735781163b306e15ff23
```